

Solutions

M-CATS

What determines how much dissolves, and what changes after it does?

Concentration / Solubility / Dissolution / Colligative properties

Recognize First

Amount

Molarity, molality, mole fraction, mass percent, ppm, dilution

Activity

Intermolecular forces, enthalpy, saturation, temperature, gas pressure

Effect

Henry's law, Raoult's law, freezing, boiling, osmosis

Warm-up: explain each choice

- 1 Which dissolves best in water: CH_4 , CH_3OH , C_6H_6 , CCl_4 ?
- 2 Equal moles in equal water: glucose or NaCl lowers T_f more?
- 3 A sealed soda warms. What happens to CO_2 solubility?
- 4 Dilute 50 mL of 2.0 M NaCl to 200 mL. Final M?

Anatomy of a solution

Solute / solvent

Solute disperses in the solvent; “aqueous” means water is the solvent.

Electrolytes

Strong electrolytes produce many ions; weak electrolytes ionize partly.

Particle count

A nonelectrolyte stays molecular. Ideal NaCl yields 2 ions per formula unit.

Concentration: choose the denominator

Molarity

$$M = \text{mol solute} / \text{L solution}$$

Molality

$$m = \text{mol solute} / \text{kg solvent}$$

Mole fraction

$$X_i = \text{mol component } i / \text{total mol}$$

Other units

Mass % = mass solute / mass solution $\times$ 100; ppm $\approx$ mg/L for dilute aqueous solutions

Normality depends on the reaction

Equivalents count the reactive units supplied per mole.

$$N = \text{equivalents} / L = M \times \text{equivalents per mole}$$

For acid–base neutralization, 0.50 M H₂SO₄ can supply up to 1.0 N H⁺. State the reaction before assigning normality.

Molarity and molality diverge

The denominator tells you which measurement is needed.

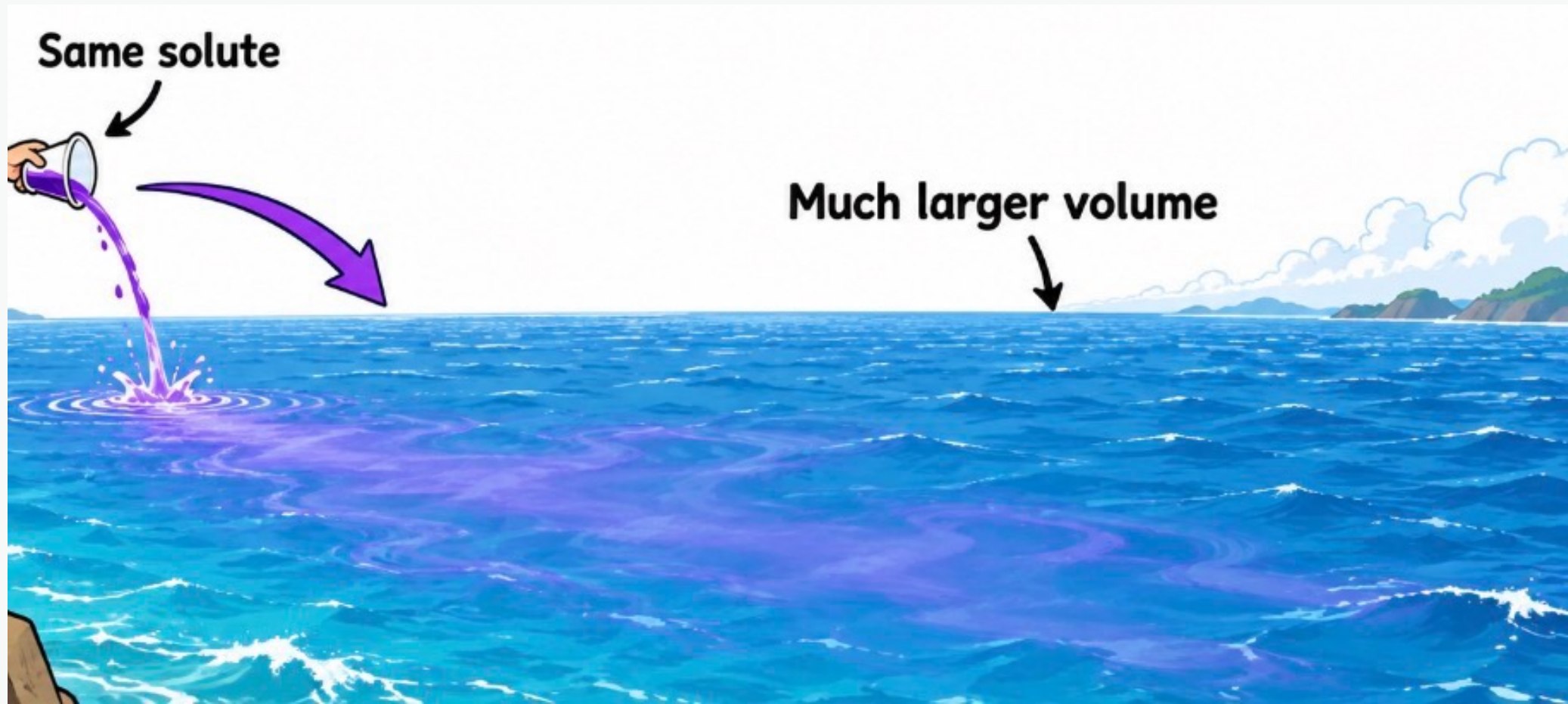
$$M = n / V_{\text{solution}} \quad m = n / m_{\text{solvent}}$$

If 1 mol is added to 0.500 kg of water, $m = 2.0 \text{ m}$. You cannot infer M without the final solution volume.

Dilution conserves solute

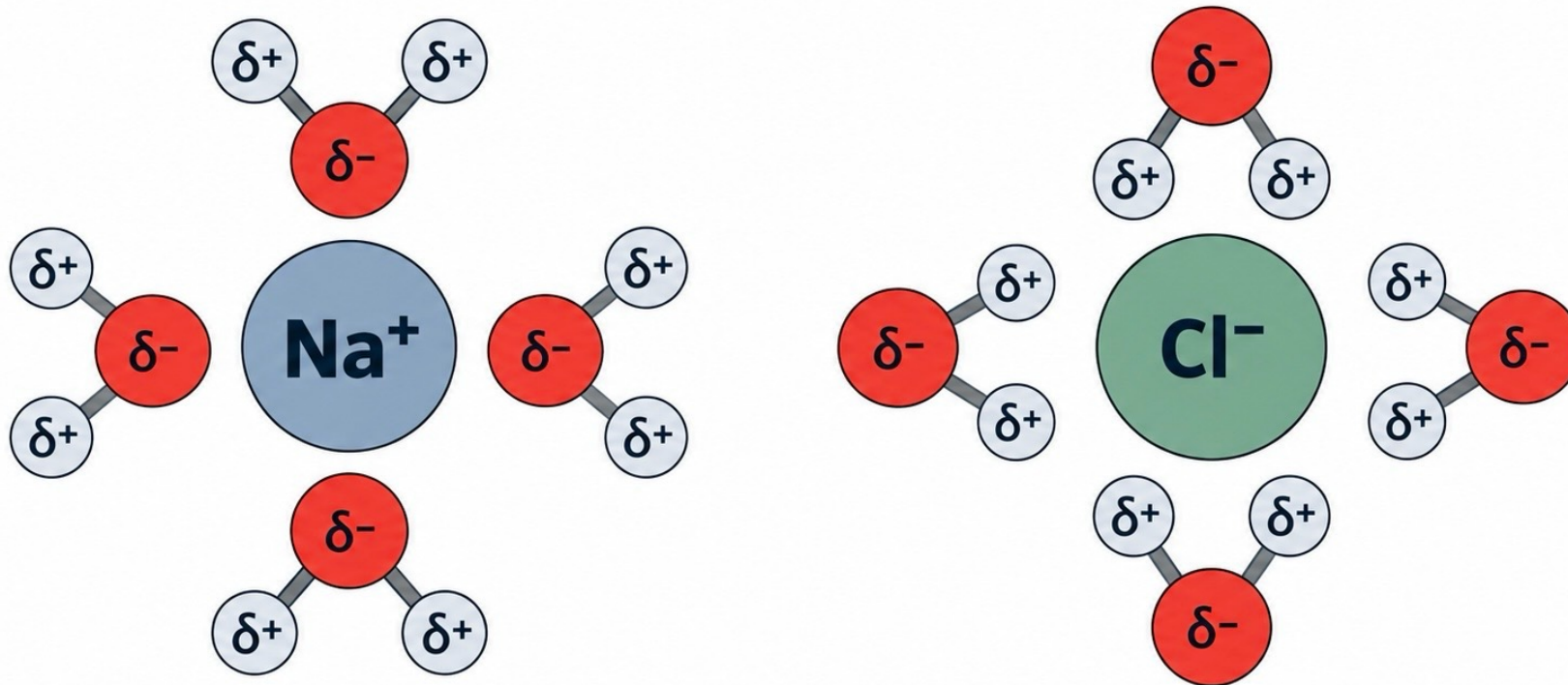
Adding solvent changes volume and concentration.

$$M_1V_1 = M_2V_2$$



Why does a solute dissolve

Form new contacts Solute-solvent interactions are strong enough, it breaks apart the solute



Dissolution has an energy balance

The heat of solution combines competing steps.

$$\Delta H_{\text{solution}} = \Delta H_{\text{separation}} + \Delta H_{\text{mixing}}$$

Concentration versus solubility

Unsaturated

Below the equilibrium maximum; more solute can dissolve.

Saturated

At the equilibrium maximum at that temperature.

Supersaturated

Above the usual equilibrium limit; crystallization may start after a disturbance.



High-yield aqueous solubility rules

Always soluble

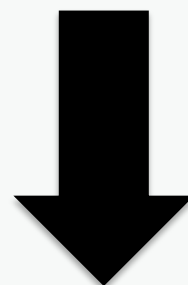
Group 1 and NH_4^+ salts; nitrate and acetate salts.

Common exceptions

Cl^- , Br^- , I^- usually dissolve except with Ag^+ , Pb^{2+} , Hg_2^{2+} .
Sulfate has Ca^{2+} , Sr^{2+} , Ba^{2+} , Pb^{2+} exceptions.

Usually insoluble

Carbonates, phosphates, sulfides, and most hydroxides unless paired with Group 1 or NH_4^+ ; some hydroxides have exceptions.

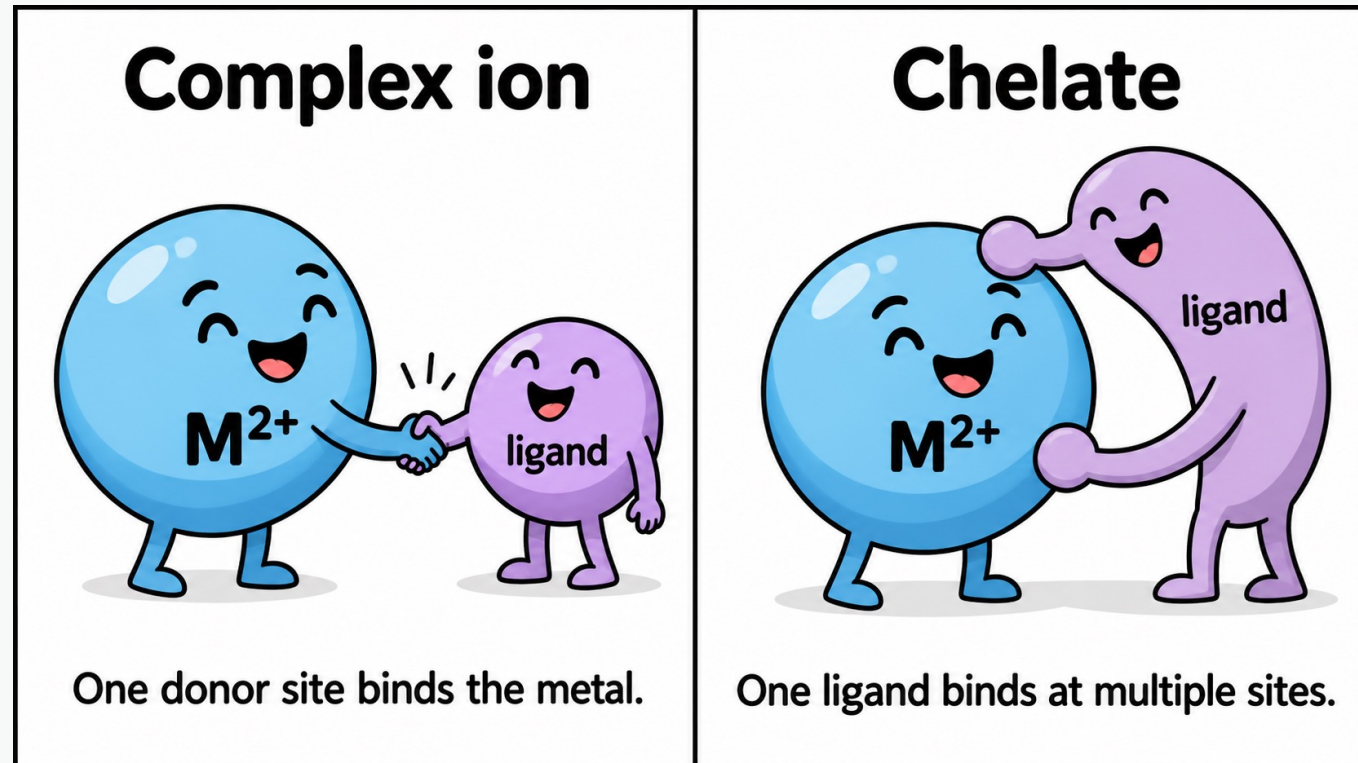


<https://chemquiz.net/ksp/>

Complex ions VS Chelation

Complex Ions: The cation (usually a metal) accepts electron pairs as a Lewis acid & binds to one ligand at each site. Can increase solubility.

Chelation: One ligand binds a metal at multiple sites; binding can shift free metal concentration and apparent solubility.



K_{sp} comes from the balanced reaction

At saturation, dissolving and precipitating occur at equal rates, so the dissolved ion concentrations stay constant. K_{sp} describes those equilibrium concentrations.

For $\text{CaF}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+} + 2\text{F}^-$, the exponent follows stoichiometry.


$$K_{sp} = [\text{Ca}^{2+}][\text{F}^-]^2$$

Ion product predicts direction

Qsp uses the current dissolved ion concentrations.

$Q_{sp} < K_{sp}$: dissolve

$Q_{sp} = K_{sp}$: saturated

$Q_{sp} > K_{sp}$: precipitate

Always account for dilution when solutions are mixed before calculating Qsp.

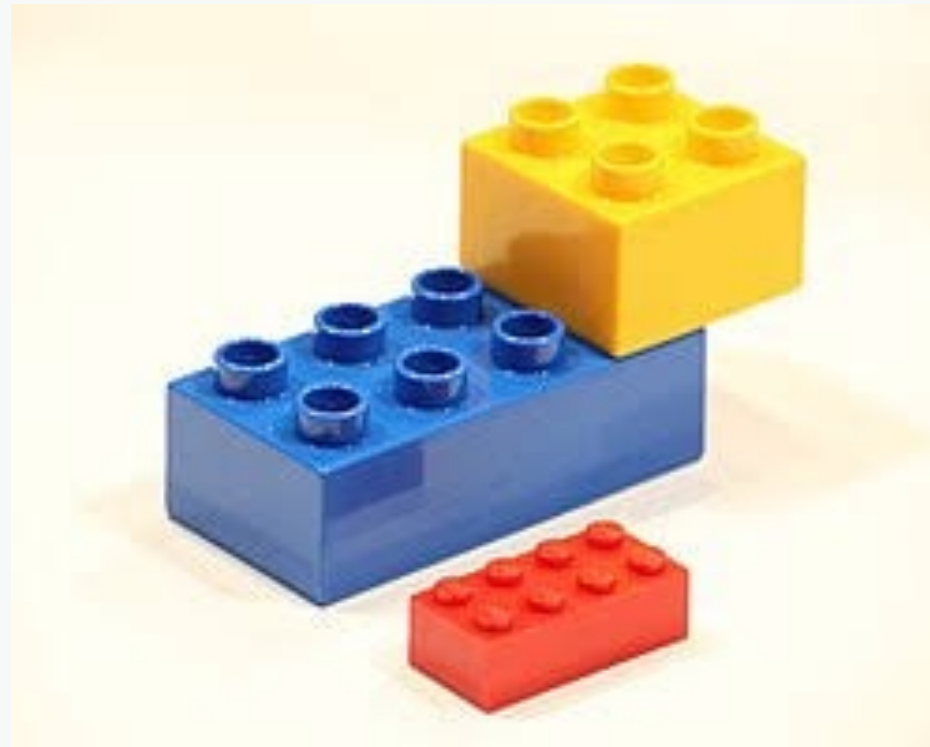
Common ions and ligands shift solubility

Add a common ion

Adding Cl^- to AgCl equilibrium reduces its molar solubility. This is the Common Ion Effect.

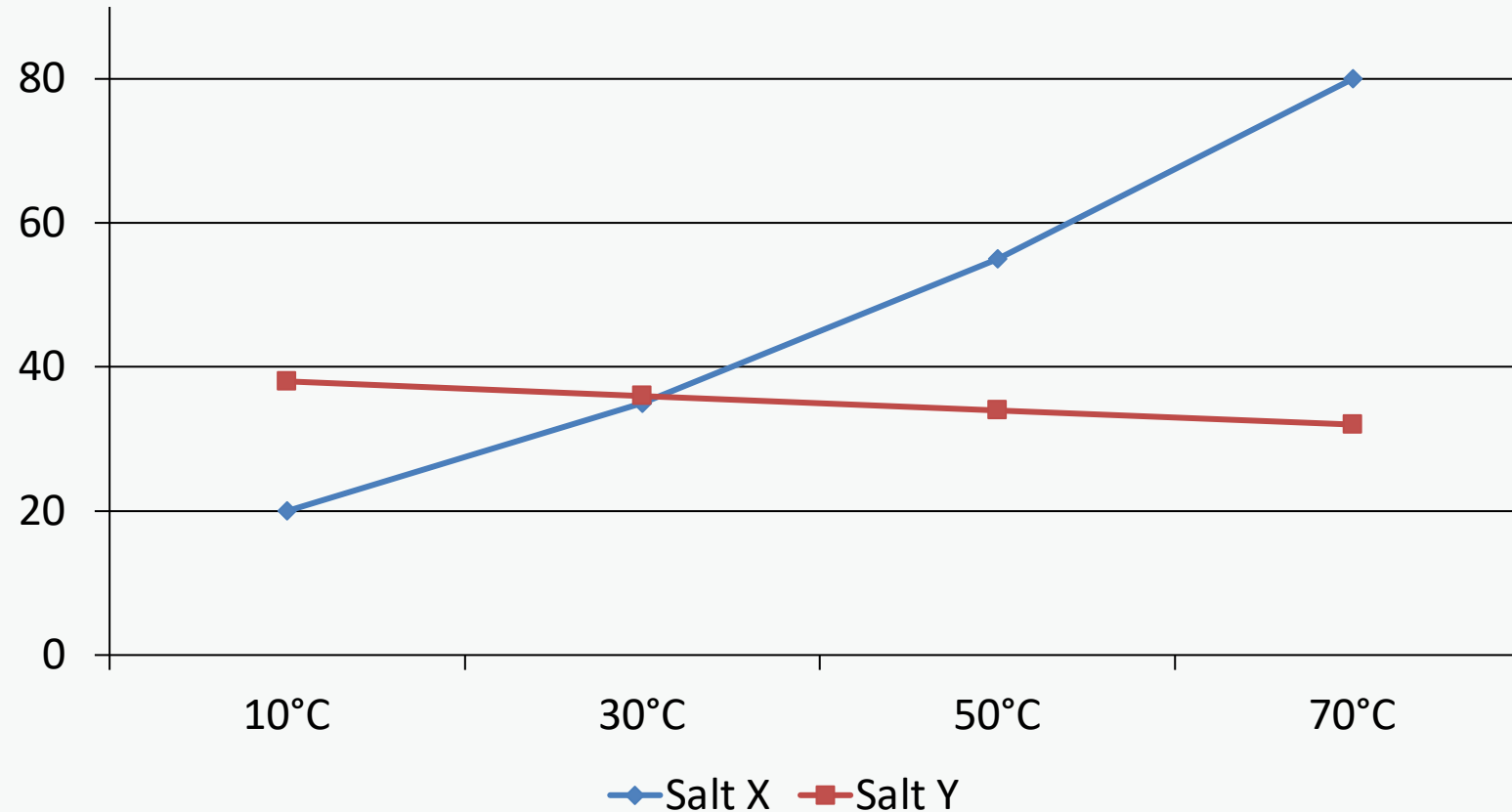
Keep K_{sp} fixed

At a specified temperature, these additions change equilibrium concentrations, not the value of K_{sp} .



Temperature affects solids differently

HYPOTHETICAL SOLUBILITY DATA: g solute per 100 g water



At 50°C:

X: 55 g

Y: 34 g

A 40 g sample of X is unsaturated. A 40 g sample of Y exceeds its solubility.

Gas solubility and pressure

At a fixed temperature, dissolved gas tracks its partial pressure.

$$C_{\text{gas}} = k_H P_{\text{gas}}$$



Doubling P_{gas} doubles equilibrium dissolved concentration under this convention. Raising temperature generally lowers gas solubility.

Raoult's law and water vapor

An ideal solution with a nonvolatile solute has less water at the liquid surface.

$$P_{\text{water}} = X_{\text{water}} P^{\circ}_{\text{water}}$$

As X_{water} falls, water vapor pressure falls. A volatile solute requires adding its own vapor pressure contribution.

One particle count, four effects

Vapor pressure

Adding a nonvolatile solute lowers the solvent vapor pressure.

Phase changes

Freezing point falls; boiling point rises.

Membrane

Osmotic pressure rises with effective particle concentration.

van't Hoff factor: count particles

Ideal dissociation predicts the number of particles per formula unit.

glucose: $i \approx 1$ NaCl: $i \approx 2$ CaCl₂: $i \approx 3$

Ion pairing and other nonideal effects can reduce the measured effective factor below the ideal prediction.

Freezing and boiling shifts

The temperature shift scales with solute particle molality.

$$\Delta T_f = i K_f m \quad \Delta T_b = i K_b m$$

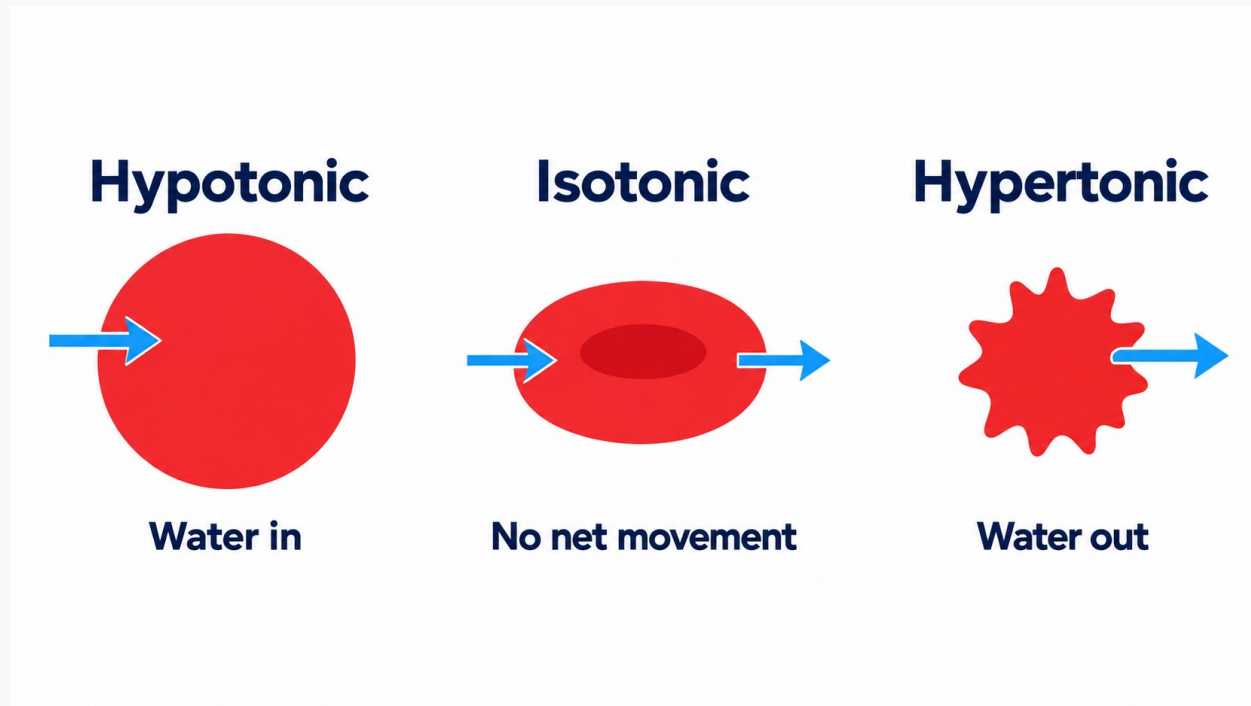
For water: $K_f = 1.86 \text{ }^\circ\text{C/m}$; $K_b = 0.512 \text{ }^\circ\text{C/m}$. Subtract ΔT_f from pure water's T_f ; add ΔT_b to pure water's T_b .

Osmosis across a membrane

A membrane lets water pass but holds solute on each side.

$$\Pi = iMRT$$

Water moves toward the side with greater effective concentration of non-penetrating solute. Cells shrink in a hypertonic extracellular solution.



Passage Question! Cryopreservation

A team studies aqueous cryoprotectants. Ice crystals can damage mammalian cells. The team prepares three solutions at 25°C and assumes complete salt dissociation unless a question states otherwise.

Solution	Solute	Molality	Ideal i
A	Glucose	0.30 m	1
B	NaCl	0.20 m	2
C	CaCl ₂	0.15 m	3

Passage B: additional experiments

Water constants: $K_f = 1.86 \text{ }^\circ\text{C/m}$; $K_b = 0.512 \text{ }^\circ\text{C/m}$.

A fourth solution contains 92.1 g glycerol (92.1 g/mol) dissolved in 500 g water. Treat glycerol as a nonvolatile nonelectrolyte.

Cells are placed in a chamber permeable to water but impermeable to its solutes. The extracellular fluid has a greater effective particle concentration.

In a separate trial, dissolved O_2 concentration rises linearly with O_2 partial pressure at constant temperature.

1. Which solution depresses freezing most?

PASSAGE QUESTION

Assuming ideal behavior, which solution has the largest freezing-point depression?

- A. Solution A
- B. Solution B
- C. Solution C
- D. All three are equal

2. Find the freezing point

PASSAGE QUESTION

What is the approximate freezing point of Solution C under the ideal assumption?

A. -0.28°C

B. -0.84°C

C. -1.67°C

D. -2.79°C

3. Explain a smaller observed shift

PASSAGE QUESTION

Solution C instead freezes at -0.70°C . Which explanation fits the observation?

- A. CaCl_2 raises water vapor pressure.
- B. The effective particle count is below three per formula unit.
- C. Dissociation raises the stated molality of CaCl_2 formula units.
- D. Dissociation decreases the number of solute particles.

4. Calculate glycerol molality

PASSAGE QUESTION

What is the molality of the glycerol solution?

- A. 0.50 m
- B. 1.0 m
- C. 2.0 m
- D. 4.0 m

5. Predict vapor pressure and boiling

PASSAGE QUESTION

Compared with pure water, the glycerol solution has which properties?

- A. Higher water vapor pressure; lower boiling point
- B. Higher water vapor pressure; higher boiling point
- C. Lower water vapor pressure; lower boiling point
- D. Lower water vapor pressure; higher boiling point

6. Determine water movement

PASSAGE QUESTION

Immediately after placement, which net movement occurs?

- A.** Water enters because intracellular osmotic pressure is greater.
- B.** Water enters because extracellular osmotic pressure is greater.
- C.** Water exits because extracellular osmotic pressure is greater.
- D.** No net flow because water concentrations must be identical.

7. Apply Henry's law

PASSAGE QUESTION

The O_2 partial pressure doubles at constant temperature. What happens to dissolved O_2 concentration at equilibrium?

- A. It halves.
- B. It remains constant.
- C. It doubles.
- D. It quadruples.

8. Change temperature

PASSAGE QUESTION

At the same O_2 partial pressure, less O_2 dissolves at higher temperature. Which statement explains this?

- A. Gas solubility in water generally decreases as temperature rises.
- B. Raoult's law makes O_2 partial pressure decrease.
- C. O_2 becomes more polar at higher temperature.
- D. Water becomes an ionic solvent at higher temperature.

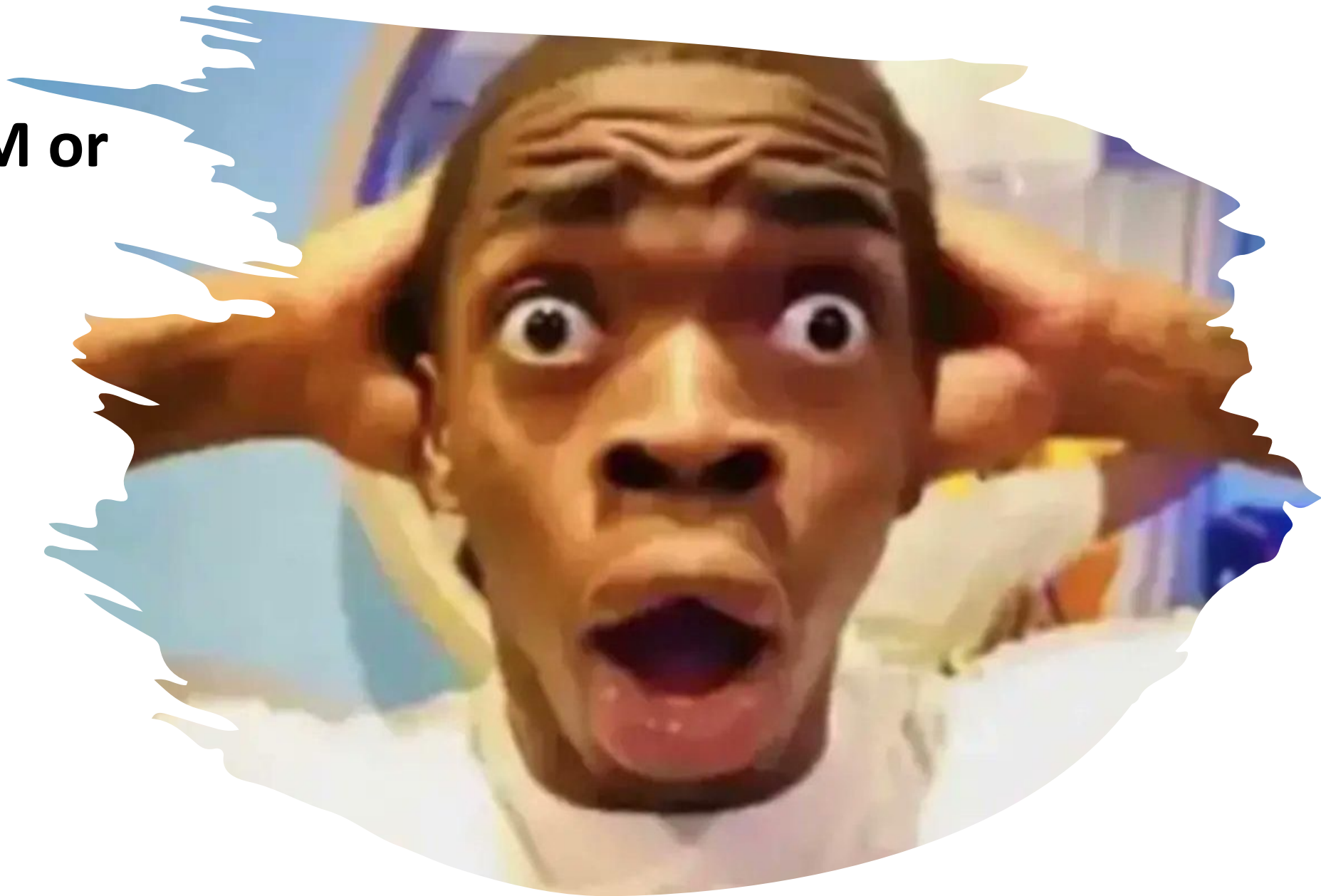
9. Dilution

PASSAGE QUESTION

A researcher takes 250 mL of 0.60 M NaCl and dilutes it to 1.00 L. Which statement is correct?

- A. 0.60 mol NaCl; boiling point below pure water
- B. 0.15 mol NaCl; freezing point above pure water
- C. 0.15 mol NaCl; water vapor pressure below pure water
- D. 0.60 mol NaCl; water vapor pressure above pure water

**WAIT.. M or
mol???**



10. Predict precipitation

CHAPTER CHECK

Equal volumes of 0.10 M AgNO_3 and 0.10 M NaCl are mixed. Assuming precipitation proceeds, what solid forms?

- A. NaNO_3
- B. AgCl
- C. AgNO_3
- D. NaCl

11. Use complex formation

CHAPTER CHECK

Excess NH_3 binds dissolved Ag^+ as $[\text{Ag}(\text{NH}_3)_2]^+$. What happens to $\text{AgCl}(\text{s})$ solubility at equilibrium?

- A. It increases as free Ag^+ is removed.
- B. It decreases because NH_3 supplies Cl^- .
- C. It is unchanged because AgCl is insoluble.
- D. AgCl converts directly to NaCl .

12. Convert units carefully

CHAPTER CHECK

For the glycerol preparation in Passage B, which pair is closest to its mass percent and glycerol mole fraction?

- A. 15.6% and 0.035
- B. 18.4% and 0.035
- C. 15.6% and 0.156
- D. 18.4% and 0.156

13. Calculate Raoult's law

CHAPTER CHECK

If pure water has a vapor pressure of 3.20 kPa at the stated temperature, what is the ideal water vapor pressure above the glycerol solution?

- A. About 3.09 kPa
- B. About 3.20 kPa
- C. About 0.11 kPa
- D. About 1.60 kPa

The relationships to retain

More particles

Lower water vapor pressure and freezing point; higher boiling point and osmotic pressure.

More gas pressure

Higher dissolved gas concentration at fixed temperature.

Higher temperature

Gas solubility usually falls; a solid's trend depends on its dissolution energetics.

Common MCAT traps

Solution volume versus solvent mass

Concentration versus equilibrium solubility

Formula units versus dissolved particles

Final volume versus solvent volume added

Gas solubility versus solid solubility

Magnitude of ΔT_f versus final freezing temperature

First identify the quantity that changes. Then choose the relationship.